COMP 6721 Course Project Proposal

Revised 10 November 2026 to match the project as carried out. The original proposal (30 September 2026) is kept in the repository history; Section 10 lists what changed.

Project title: Mutational Signature Detective: Estimating Carcinogen Exposure from Tumor Mutation Patterns

Topic mapping: Custom topic, framed as ML-3/ML-4 (imbalanced, explainable signal decomposition) applied to genomics. Instructor approval requested.

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Repository: github.com/ralegionc/COMP6721-F26-PROJECT

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1. Problem statement

Carcinogens (tobacco, UV light) and internal processes (ageing, DNA-repair defects) leave characteristic mutation patterns in tumor DNA, called mutational signatures. Given a tumor's mutation catalog (counts across the 96 trinucleotide contexts, the standard SBS representation), we estimate exposure weights over known reference signatures (for example SBS4 = tobacco, SBS7a to SBS7d = UV, SBS1 and SBS5 = ageing). The estimate for one tumor is its exposure vector: non-negative shares, one per candidate signature, summing to 1.

We do not claim clinical or legal proof of exposure; some signatures (ageing) are not external exposures at all. The aim is to estimate and compare attribution methods, and to characterize when attribution is reliable.

Research questions:

1. How accurately do NNLS, elastic-net NNLS, a multinomial maximum-likelihood fit and a small neural network recover known exposures?
2. At what mutation count does accuracy break down? We define the breakdown point N* as the smallest mutation count from which the median cosine similarity between true and estimated exposures stays at or above 0.90.
3. How much does accuracy drop when a tumor carries a signature that is missing from the method's candidate list? (Optional extension: does performance hold under a cohort shift?)
4. Is the estimated tobacco-signature (SBS4) weight higher in smokers with lung adenocarcinoma than in lifelong non-smokers?

2. Data and resources

- **Reference signatures:** COSMIC SBS v3.4, GRCh38, exome-normalized (86 signatures), from the SigProfilerAssignment distribution. For each cancer type we fit only the signatures that Alexandrov et al. (2020) attributed in at least 5% of that type's TCGA exomes, excluding the 19 signatures COSMIC labels as sequencing artifacts: 6 for LUAD, 7 for SKCM and 8 for COAD.
- **TCGA (GDC Data Portal), open-access "Masked Somatic Mutation" MAF files,** filtered to MuTect2-supported calls, GRCh38 reference genome. Three cancer types chosen for distinct dominant signatures: lung adenocarcinoma (LUAD, tobacco), skin melanoma (SKCM, UV) and colon adenocarcinoma (COAD, ageing and mismatch repair). 1,547 files downloaded and checksum-verified; 1,536 tumor samples from 1,451 patients after filtering.
- **TCGA clinical metadata** (smoking status, pack-years) for real-data checks only, not ground truth.
- **PCAWG (optional):** used only if the cohort-shift extension goes ahead, in which case both cohorts switch to the MC3 public MAF so that they share the GRCh37 build.

All sources are public, require no access request or license agreement, and no restricted data is uploaded to Git.

Label caveat: real tumors have no confirmed ground-truth exposure weights, and clinical fields are imperfect proxies. Primary validation therefore uses simulation; real data gives directional checks only.

3. Baseline and methods

- **Baseline B0 (cohort mean):** predicts the average training exposure for every tumor, the attribution equivalent of always predicting the majority class. Every method must beat it.
- **M0, NNLS:** non-negative least squares against the fixed signature matrix (the standard classical approach).
- **M1, elastic-net NNLS:** NNLS with L1 and L2 penalties for low-mutation robustness; the penalty strength and L1 share are tuned on validation data only.
- **M2, multinomial maximum likelihood:** fitted with the expectation-maximization (EM) algorithm, with bootstrap intervals giving uncertainty over exposure weights. This replaces the Bayesian fit in the original proposal; a Bayesian (MCMC) fit remains optional.
- **M3, small neural network** (multilayer perceptron) trained on simulated tumors, as a core method, expected to make different errors than the least-squares family.
- **Analysis:** a signature-by-signature confusion matrix from simulation (true against attributed dominant signature), and how errors change with mutation count.

4. Evaluation plan

- **Split:** patients are split 60/20/20 into training, validation and test sets by 12-character TCGA patient ID, stratified by cancer type, before the simulator is calibrated. The split and all simulated sets are saved once (seed 6721, SHA-256 fingerprints) and every method reads the same files.

- **Primary (simulation):** synthetic tumors with known signature mixtures, calibrated to the mixture and mutation-burden statistics of the real training patients only, with extra count noise and per-tumor signature distortion. Test tumors are generated at fixed mutation counts from 25 to 5,000.
- **Metrics:** cosine similarity and mean absolute error on truly present signatures; presence detection (weight of at least 5%) with per-signature F1 and PR-AUC; the confusion matrix; the breakdown point N*. Accuracy is reported only to show how much an "always absent" prediction would inflate it.
- **Statistical comparison:** paired Wilcoxon signed-rank tests between methods on the same test tumors, with Holm's correction for multiple comparisons.
- **Unseen signatures (E3):** simulate tumors containing a signature left out of the candidate list (for example SBS10a/b or SBS17a) and measure the loss in accuracy.
- **Real data (statistical test):** in TCGA lung adenocarcinoma, test whether the estimated SBS4 weight is higher in current or reformed smokers than in lifelong non-smokers (one-sided Mann-Whitney U), with negative controls (UV signatures in lung tumors, SBS4 in melanoma). This is a falsifiable directional check, not an accuracy claim.
- **Optional:** cohort shift by tissue source site or PCAWG; comparison with SigProfilerAssignment or deconstructSigs on simulated data.

5. Division of work

- **Anas Farooqui:** data acquisition and preprocessing (TCGA download and checksums, MAF parsing, context extraction, caller filtering, clinical labels, signature subsets, patient split, storage) and the real-data checks.
- **Hugo Xu:** simulation framework; classical and likelihood methods (B0, M0, M1, M2); statistical tests.
- **Bao Tran Nguyen:** neural network method (M3); evaluation, error analysis and figures.
- All members share the report, reproducibility package and presentation, and review each other's pull requests.

6. Plan

Weeks	Work	Status (10 Nov)
3 to 8	Download TCGA data; reference files and signature subsets; context-extraction pipeline; patient split; simulator; B0, M0, M1, M2; first accuracy-against-mutation-count results; Git repository with README	Done
9	Statistical tests in the pipeline; M1 penalty grid widened and tuned on validation data	Done
9 to 10	Neural network (M3) trained and tuned on simulated tumors	In progress
10 to 11	E3 unseen signatures; extended real-data checks with a count-dependent presence threshold; M2 interval coverage	Planned

12	Error analysis (confusion between similar signatures, low-count tumors); final figures	Planned
13	Report, reproducibility check from a clean clone, presentation	Planned

7. Risks and fallbacks

1. **Overreaching exposure claims.** Mitigation: report outputs as estimated contributions, not diagnoses; note that ageing is not an exposure.
2. **No real ground truth.** Mitigation: simulation is the primary validation; real data is a directional check with negative controls.
3. **Simulator too easy.** Simulated tumors follow the methods' own model, which inflates accuracy. Mitigation: extra noise and signature distortion in the simulator, and E3 reported next to the main results.
4. **Low-mutation tumors are hard cases.** Mitigation: treat this as an analyzed, expected result (the breakdown point), not a bug.
5. **Neural network learns the simulator.** Mitigation: compare M3 with M0 on the real-data checks too, not only on simulations.
6. **Scope.** Mitigation: the guaranteed minimum is B0 to M3 on the three cancer types with simulation validation and the smoking test; cohort shift, PCAWG and the published-tool comparison are cut first if behind schedule.

8. Limitations

- Signatures indicate mutational processes, not confirmed exposures; ageing is not external.
- Real-data validation is indirect (clinical proxies), not ground truth.
- Attribution is unreliable at low mutation counts; we characterize rather than hide this.
- Open-access MAFs remove some low-frequency variants, lowering counts in low-burden tumors; rare signatures (under 5% of a type's tumors) are not candidates.
- This is a research prototype for hypothesis generation, not a diagnostic tool.

9. Ethics and integrity

Only open-access public data is used; no controlled-access data is accessed or uploaded. External data, tools and papers (COSMIC, TCGA/GDC, Alexandrov et al. 2020, SigProfilerAssignment, and PCAWG and deconstructSigs if used) will be cited.

10. Changes since the original proposal

- **Cancer types:** three (LUAD, SKCM, COAD) instead of five or six; they cover the tobacco, UV and ageing signatures, and three types fit the timeline.
- **PCAWG cross-cohort test:** moved from a core test to an optional extension, because adding PCAWG requires switching both cohorts to GRCh37. The unseen-signature

experiment (E3) takes its place as the robustness test.

- **Bayesian method:** replaced by a multinomial maximum-likelihood fit (EM) with bootstrap intervals, which also gives uncertainty over the weights; a Bayesian fit remains optional.
- **Added:** the cohort-mean baseline B0, presence F1 and PR-AUC, the breakdown point N*, paired Wilcoxon tests with Holm correction, and negative controls for the smoking test.
- **Published-tool benchmark** (SigProfilerAssignment, deconstructSigs): moved to optional.
- **Division of work:** members named; the evaluation statistics moved from Member 3 to Member 2.